

Mission-Adaptive Reconfigurable Battery Management System for Autonomous Rovers

B.Tech Capstone Project Proposal

Project Domain	Embedded Systems / Power Electronics / Autonomous Vehicles
Platform	Rover / Unmanned Ground Vehicle (UGV)
Battery Architecture	6-Cell Li-Ion with Three-Switch Reconfigurable Topology
Key Innovation	SOC and SOA driven real-time topology reconfiguration
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1. Introduction and Motivation

The rapid growth of autonomous vehicles — rovers, drones, and underwater vehicles — has placed increasing demands on battery systems. These platforms require reliable, efficient, and intelligent power management to maximize mission duration and ensure operational safety. The Battery Management System (BMS) is the critical subsystem responsible for monitoring battery health, controlling charge and discharge processes, and protecting cells from unsafe operating conditions.

Traditional BMS designs treat the battery pack as a fixed electrical configuration — cells are permanently wired in series or a fixed series-parallel arrangement, and the BMS functions purely as a passive watchdog. It monitors parameters and disconnects the entire pack when any single cell exceeds its safe operating limits. While this approach provides basic protection, it leaves significant energy unused, offers no fault tolerance during missions, and cannot adapt to varying load demands.

Recent research in reconfigurable battery systems (RBS) has demonstrated that dynamically changing the electrical connections between cells — switching between series, parallel, and bypass configurations — can dramatically improve energy utilization, extend pack lifetime, and enable continued operation even when individual cells fail. However, this technology has been explored primarily at a theoretical level for large-scale energy storage systems. No study has implemented a mission-adaptive reconfigurable BMS on a small autonomous vehicle platform.

This project bridges that gap by designing, building, and validating a reconfigurable BMS for an autonomous rover. The system uses State of Charge (SOC) and Safe Operating Area (SOA) as real-time decision inputs to dynamically alter the battery pack topology — enabling fault tolerance, wear leveling, and maximum energy extraction from every cell in the pack.

2. Problems in Traditional Battery Management Systems

2.1 The Weakest Cell Problem

In a fixed series-connected battery pack, the same current flows through every cell. The discharge process is terminated the moment any single cell reaches its undervoltage cutoff threshold. This is a direct consequence of Kirchhoff's current law — in a series circuit, current cannot flow through cell 2 without simultaneously flowing through cell 1. If cell 1 reaches its minimum safe voltage while cells 2 through 6 still have 30–50% charge remaining, the entire pack must shut down. The usable energy remaining in the healthy cells is completely stranded and inaccessible.

This problem is mathematically described by the discharge cutoff equation from reconfigurable battery system literature: $V_{\text{cut-discharge}} = V_t^i - \min\{V_t^1, V_t^2, \dots, V_t^M\} + V_{\text{fchar}}$. The entire string's cutoff is governed by the weakest cell. After 100+ charge-discharge cycles, cell-to-cell SOC drift of 5–15% is common due to manufacturing variance, uneven thermal exposure, and differential aging. Even a 10% imbalance means 10% of the pack's usable energy is wasted every cycle.

2.2 Binary Fault Response

When a traditional BMS detects a fault condition — overvoltage, undervoltage, overcurrent, overtemperature, or abnormal internal resistance — it has only one available response: disconnect the entire battery pack. For an autonomous rover mid-mission on rough terrain, this means immediate and total loss of power. There is no graceful degradation, no reduced-capacity operation, and no ability to isolate the problematic cell while continuing on the remaining healthy cells. One cell's failure becomes the entire system's failure.

2.3 No Adaptation to Load Conditions

A rover traversing varied terrain experiences dramatically different power demands. Flat surface cruising requires moderate, steady current. Climbing an incline demands high current bursts. Turning on loose ground creates transient current spikes. A traditional BMS provides the same fixed voltage-current characteristic regardless of the operational phase. The pack configuration cannot adapt to what the vehicle actually needs. This forces engineers to oversize the battery pack for worst-case current scenarios, adding unnecessary weight and cost for the 90% of the mission that does not require peak power.

2.4 Slow and Wasteful Cell Balancing

Traditional passive balancing dissipates excess charge from higher-SOC cells through resistors, converting usable energy into waste heat. Active balancing using external inductor or capacitor circuits adds cost, board space, and additional failure points. Both approaches operate at balancing currents of milliamps against load currents of amps — they are fundamentally too slow to address imbalances during active operation and can only make marginal corrections during idle or charging periods.

2.5 Underutilized State Information

Traditional BMS implementations estimate SOC and SOH to display battery gauge information to the user and to flag eventual pack replacement. These valuable state estimates are used for monitoring and display purposes only — they do not feed into any active decision-making loop that changes how the battery pack operates. The intelligence exists in the data, but the system lacks the mechanism to act on it.

3. Proposed Solution: Mission-Adaptive Reconfigurable BMS

3.1 Core Concept

We propose a Battery Management System that replaces the fixed battery topology with a dynamically switchable one. Instead of cells permanently wired together, each cell is connected through a set of three N-channel MOSFETs — a series switch, a parallel switch, and a bypass switch — that allow the BMS to change how cells are electrically connected in real time, under load, during a live mission.

The system uses 6 individual Li-ion cells (21700 format), each with its own MOSFET switching circuit, current sensing, voltage sensing, and temperature monitoring. An STM32 microcontroller runs the reconfiguration algorithm and handles all monitoring, protection, and communication functions.

The fundamental innovation is that SOC and SOA are not merely monitored and displayed — they serve as real-time inputs to an active control loop that dynamically alters the physical electrical configuration of the battery pack. The BMS does not just protect; it reconfigures around problems to maintain operational continuity.

3.2 Three-Switch Topology

The reconfigurable topology employs three MOSFET switches per cell, enabling four connection states for each cell: series connection (cell contributes voltage in the string), parallel connection (cell shares current load with adjacent cells), bypass (cell is electrically isolated from the active circuit), and hybrid series-parallel (combinations of the above across the pack). This is the three-switch topology identified in the literature as offering the best balance of reconfiguration flexibility, cost, and control complexity for small-to-medium scale applications.

Cell Connection States:

State	Series Switch	Parallel Switch	Bypass Switch	Function
Series	CLOSED	OPEN	OPEN	Cell adds voltage to string
Parallel	OPEN	CLOSED	OPEN	Cell shares current with adjacent cell
Bypass	OPEN	OPEN	CLOSED	Cell isolated, current bypasses it
Off	OPEN	OPEN	OPEN	Cell fully disconnected

Table 1: MOSFET switch states for each cell connection configuration

3.3 Nominal Configuration

The rover's motor drivers and electronics expect a stable voltage, typically in the 11.1V range. The nominal battery configuration is 3S2P — two parallel strings of three series-connected cells. This provides 11.1V nominal at double the single-cell current capacity. The

reconfiguration system maintains this target voltage while dynamically managing which cells participate in each string, how current is distributed, and which cells are bypassed or rested.

3.4 SOC-Driven Reconfiguration

The SOC of each cell is estimated in real time using an Extended Kalman Filter (EKF) with coulomb counting as a secondary method. SOC values directly drive reconfiguration decisions. Cells with higher SOC are prioritized for active duty. When a cell's SOC drops below a defined threshold (e.g., 20%) while other cells remain above 35%, the low-SOC cell is bypassed and the remaining cells are reconfigured to maintain the closest achievable output voltage. This prevents the weakest cell from forcing a full pack shutdown and extracts usable energy from every cell individually.

3.5 SOA-Driven Reconfiguration

The Safe Operating Area defines the boundaries within which a cell can operate without risk of damage or safety hazard. SOA is defined by three parameters: voltage limits (2.5V minimum, 4.2V maximum), current limits (continuous discharge rating of the cell), and temperature limits (typically 0–60°C for discharge). When any cell's operating point approaches or crosses an SOA boundary, the BMS immediately reconfigures the topology to address the issue — bypassing an overheating cell, redistributing current away from an overstressed cell, or isolating a cell showing voltage anomalies. This is fundamentally different from a traditional BMS, which would shut down the entire pack in response to the same conditions.

3.6 Wear Leveling Through Topology Rotation

Over extended operation, the system tracks cumulative energy delivered by each cell. Cells that have delivered disproportionately more energy are periodically bypassed while fresher cells take over their position in the active configuration. This rotation equalizes aging and degradation across all cells, extending the overall pack lifetime. Unlike traditional cell balancing that wastes energy through resistors, this approach balances cells by managing their workload — no energy is dissipated unnecessarily.

3.7 Standard Protection Layer

All conventional BMS safety features are retained as a foundational protection layer beneath the reconfiguration logic. These include per-cell overvoltage and undervoltage protection, pack-level overcurrent protection, short circuit detection and isolation, per-cell overtemperature and undertemperature monitoring, and pre-charge/pre-discharge current ramping. The critical difference is that these protections trigger reconfiguration as a first response. Full pack shutdown is the last resort, invoked only when reconfiguration cannot resolve the issue.

4. System Architecture

4.1 Hardware Architecture

The hardware system consists of three primary subsystems: the battery cell array with individual MOSFET switching circuits, the sensing and measurement subsystem, and the central control unit.

Component	Specification	Quantity	Function
Battery Cells	Samsung INR21700-50E 3.7V, 5Ah, 10A cont.	6	Energy storage
Series Switch	N-MOSFET, low RDS(on) ~2.5mΩ, 40V, 120A	6	Series cell connection
Parallel Switch	N-MOSFET, low RDS(on) ~2mΩ, 80V, 170A	6	Parallel cell connection
Bypass Switch	N-MOSFET, low RDS(on) ~2.5mΩ, 40V, 120A	6	Cell bypass/isolation
Gate Drivers	Half-bridge gate driver IC	9	MOSFET control signals
Current Sensor	INA226 (I2C interface)	3	Per-string current + voltage
Temp Sensors	NTC 10K Thermistor	6	Per-cell temperature
MCU	STM32F446 or ESP32	1	Central control and algorithm
CAN Transceiver	MCP2551 or SN65HVD230	1	Vehicle communication

Table 2: Hardware component specifications

4.2 Software Architecture

The firmware is structured as a real-time loop executing at 100ms intervals with the following pipeline: sensor data acquisition (voltage, current, temperature for all 6 cells), state estimation (per-cell SOC via EKF, SOA boundary checking), reconfiguration decision engine (rule-based heuristic algorithm evaluating all trigger conditions), MOSFET command execution (safe switching sequence with transition management), and data logging and communication (CAN bus to vehicle controller, optional telemetry).

4.3 Power Budget of BMS Overhead

Component	Power Consumption
MCU (STM32F446)	100–150 mW
Sensors (INA226 x3, NTC x6)	20–50 mW
Gate Drivers (9 channels)	30–50 mW
CAN Transceiver	10–20 mW

MOSFET Conduction Loss (I^2R at 5A, 3 switches)	~300 mW
Total BMS Overhead	~500–600 mW

Table 3: BMS power overhead budget

Over a 1-hour mission, the BMS consumes approximately 0.5–0.6 Wh. In contrast, the energy recovered by preventing stranded capacity in imbalanced cells is typically 15–20 Wh for a 6-cell pack with 10% cell imbalance. This represents a return of approximately 25–30x on the energy invested in BMS operation.

5. Reconfiguration Decision Logic

5.1 Trigger Conditions

The reconfiguration algorithm evaluates three categories of triggers at every control cycle (100ms). Any single trigger is sufficient to initiate reconfiguration. Multiple simultaneous triggers are resolved by priority, with safety-critical SOA triggers taking precedence over efficiency-driven SOC triggers.

Trigger 1 — Current Approaching Cell Limit:

Per-string current is monitored via INA226 sensors. When current in any parallel string exceeds 80% of the cell's continuous discharge rating (e.g., 8A for a 10A-rated cell), the system reconfigures to bring additional cells into parallel paths, distributing the load across more strings. For example, if total load jumps from 12A (6A per string in 2-string parallel) to 18A (9A per string, exceeding the 8A threshold), the BMS reconfigures to create three parallel paths, reducing per-string current to 6A.

Trigger 2 — Temperature Exceeding SOA:

Per-cell temperature is monitored via NTC thermistors. If any cell's temperature exceeds the warning threshold (e.g., 50°C) or shows a rapid rise exceeding 2°C/min, the BMS proactively bypasses that cell and reconfigures the remaining cells. This prevents thermal runaway escalation and allows the isolated cell to cool passively while the rover continues operating.

Trigger 3 — Voltage Sag Under Load:

Cell terminal voltage is monitored under load. If a cell's voltage drops below a minimum under-load threshold (e.g., 3.0V) while its SOC estimate indicates it should still have capacity (SOC > 20%), this indicates abnormally high internal resistance — a sign of cell degradation. The BMS bypasses the degraded cell and flags it for inspection.

Trigger 4 — SOC Imbalance:

When the difference between the highest and lowest per-cell SOC exceeds a threshold (e.g., 15%), the BMS begins SOC-based wear leveling. The lowest-SOC cell is bypassed and rested while higher-SOC cells continue serving the load. When the active cells' SOC drops to match the resting cell, it is brought back into the active configuration. This equalizes SOC without wasting energy through balancing resistors.

5.2 Switching Transition Sequence

Reconfiguration cannot happen instantaneously under full load without risking current spikes, shoot-through, or arcing. The firmware implements a safe transition sequence: (1) Reduce motor PWM duty cycle to lower load current over 2–3ms. (2) Open switches on the cells being reconfigured (break-before-make). (3) Close switches for the new configuration. (4) Ramp motor PWM back to operational level over 2–3ms. The total reconfiguration window is approximately 5–10ms. The rover experiences a brief, imperceptible hesitation in drive torque.

5.3 Algorithm Pseudocode

```
EVERY 100ms:

    READ I_string[i] for each parallel path
    READ T_cell[i] for each cell
    READ V_cell[i] for each cell under load
    UPDATE SOC[i] via Extended Kalman Filter

    // SOA Priority Checks (Safety-Critical)
    IF any T_cell[i] > T_max OR T_rate > 2°C/min:
        → BYPASS overheating cell, RECONFIGURE remaining
    IF any V_cell[i] < V_min while SOC[i] > 20%:
        → BYPASS degraded cell, FLAG for inspection
    IF any I_string[i] > 80% of I_rated:
        → ADD cells to parallel paths, REDISTRIBUTE current

    // SOC Optimization Checks (Efficiency)
    IF max(SOC) - min(SOC) > 15%:
        → BYPASS lowest-SOC cell, WEAR LEVEL
    IF any SOC[i] < SOC_min_threshold:
        → BYPASS depleted cell, CONTINUE on remaining

    // Load Normalization
    IF total load returns below normal threshold:
        → RESTORE nominal configuration
        → REST previously overworked cells
```

Table 4: Reconfiguration decision algorithm pseudocode

6. Operational Scenarios and Test Cases

The following scenarios define the specific conditions under which the reconfigurable BMS will be tested and validated on the rover platform. Each scenario represents a realistic operating condition that demonstrates a distinct advantage of reconfiguration over traditional BMS behavior.

#	Scenario	Condition	Traditional BMS Response	Our BMS Response	Measurable Outcome
1	Normal cruise	Flat terrain, moderate load	Fixed config, no action	Nominal 3S2P, all 6 cells active	Baseline performance
2	Steep incline	Current spikes above 80% cell rating	No adaptation, cells overstress	Add parallel paths, redistribute current across more cells	Reduced per-cell current, lower temperature rise
3	Single cell overheat	One cell temp exceeds 50°C	Full pack shutdown	Bypass hot cell, continue on remaining 5 cells	Mission continues at ~83% capacity
4	SOC imbalance	One cell at 20% others at 40%	Full pack shutdown at weakest cell	Bypass low cell, extract remaining energy from others	15–30% more runtime from same pack
5	Cell fault (resistance spike)	One cell shows abnormal voltage sag under load	Full pack shutdown	Bypass faulty cell, isolate and flag, continue mission	Fault tolerance, no mission failure
6	Extended operation	Uneven wear accumulating over 30+ min	No action, uneven aging	Rotate active/resting cells to equalize wear	More uniform cell degradation over lifetime
7	Charging	Pack plugged in for recharge	Fixed series charging	All cells parallel for fast, balanced charging	Faster charge, automatic equalization

Table 5: Complete test scenario matrix with expected outcomes

6.1 Demo Setup

For the capstone demonstration, the rover will be equipped with a small ramp or incline surface. The test sequence will proceed as follows: (1) Flat surface cruise demonstrating nominal 3S2P operation. (2) Ramp climb showing current redistribution via parallel reconfiguration. (3) Simulated cell fault by manually heating one cell with a heat gun to trigger SOA-based bypass. (4) Pre-imbalanced SOC test where one cell is deliberately partially discharged before the demo to show SOC-based bypass and extended runtime. Throughout the demo, a laptop connected via CAN/UART will display real-time per-cell SOC, temperature, current, and active topology configuration.

6.2 Edge Cases

Edge Case	System Behavior
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Two cells fault simultaneously	Both bypassed; if remaining 4 cells cannot maintain minimum required voltage, controlled shutdown with warning
All cells below 20% SOC	Low-power mode: reduce motor PWM ceiling, alert vehicle controller, begin return-to-base
MOSFET switch failure (stuck open)	Detected via voltage sensing; cell excluded from all future configurations; system continues on remaining cells
MOSFET switch failure (stuck closed)	Detected via unexpected current path; adjacent switches opened to isolate; emergency shutdown if isolation fails
Rapid successive load transients	Reconfiguration cooldown timer prevents oscillation; minimum 500ms between topology changes
Communication loss with vehicle	BMS operates autonomously on last known mission profile; defaults to safety-priority mode
Ambient temperature below 0°C	Under-temperature protection activated; charging disabled; discharge allowed at reduced current only

Table 6: Edge case handling

7. Advantages Over Traditional BMS

7.1 Quantitative Advantages

Metric	Traditional BMS	Our Reconfigurable BMS	Improvement
Usable energy extraction	70–85% (limited by weakest cell)	90–98% (per-cell extraction)	15–30% more runtime
Fault response	Full pack shutdown (0% output)	Bypass faulty cell (83% output with 5/6 cells)	Mission continuity
Cell balancing method	Passive resistor (wastes energy as heat)	Topology-based workload rotation (zero waste)	No balancing energy loss
Balancing hardware	External resistors, capacitors, or inductors	Uses existing MOSFET switches (no extra HW)	Reduced BOM cost for balancing
Pack lifetime	Weakest cell degrades fastest (positive feedback)	Wear leveling equalizes degradation (negative feedback)	20–40% longer pack life
Load adaptation	Fixed configuration regardless of demand	Dynamic parallel paths for high-current demand	Reduced per-cell stress
Degraded operation	Not supported; failed pack = replacement	Operates on 4–5 cells at reduced capacity	Extends usable pack life

Table 7: Quantitative comparison with traditional BMS

7.2 Energy Recovery Analysis

Consider a 6-cell pack with each cell at 3.7V nominal and 5Ah capacity, giving a total pack energy of 111 Wh. In a traditional fixed series pack with 10% SOC imbalance between cells, approximately 10–15% of the pack energy (11–17 Wh) is stranded in healthy cells when the weakest cell triggers shutdown. The BMS overhead consumes approximately 0.5–0.6 Wh over a 1-hour mission. The net energy recovery is therefore 10–16 Wh — a return of 17–27x on the energy invested in BMS operation. Even under the conservative assumption of 5% cell imbalance, the net recovery is 5–8 Wh, still a highly favorable 8–13x return.

7.3 Industry Alignment

This architecture mirrors the direction in which the battery industry is moving. Tesla implements individual cell monitoring across thousands of cells. BYD and CATL use modular architectures with module-level management. The growing academic interest in reconfigurable battery systems (70+ papers referenced in the 2026 Tang et al. survey) confirms this is an active and relevant research direction. The project demonstrates this emerging industrial trend at a proof-of-concept scale with a novel application on an autonomous rover platform.

8. Constraints and Limitations

8.1 Scalability

Per-cell switching with three MOSFETs per cell is feasible for our 6-cell proof-of-concept but does not scale linearly to large packs. A 4,000-cell EV pack would require 12,000 MOSFETs, making per-cell reconfiguration impractical at that scale. The scalable solution is hierarchical: cells are grouped into fixed modules of 6–8 cells each, and reconfiguration is applied at the module level. Our project validates the concept and algorithms at cell level; the same logic applies at module level in a production system.

8.2 MOSFET Conduction Losses

Each active current path passes through 2–3 MOSFETs, each contributing I^2R losses. At 5A through a $4\text{m}\Omega$ MOSFET, each switch dissipates 100mW. Three switches in the path means 300mW of conduction loss. While this is small relative to the pack's output power ($\sim 55\text{W}$ at nominal), it is nonzero and represents a permanent efficiency overhead that a fixed-topology pack does not have. Low-RDS(on) MOSFETs are selected to minimize this impact.

8.3 Switching Transients

Reconfiguration during active load introduces brief transient disruptions (5–10ms). While imperceptible for a rover's drive motors, this could be problematic for sensitive electronics. The system mitigates this by reducing load before switching and ramping back up after. Additionally, a reconfiguration cooldown timer (minimum 500ms between topology changes) prevents rapid oscillation between configurations.

8.4 Complexity and Failure Modes

The addition of 18 MOSFETs, gate drivers, and associated control logic introduces new potential failure modes that do not exist in a traditional BMS. A stuck-closed MOSFET could create an unintended current path. A stuck-open MOSFET could permanently disconnect a cell. The firmware must include MOSFET health monitoring (verifying expected voltage/current after each switching operation) and safe fallback states. This added complexity is the fundamental trade-off for the reconfiguration capability.

8.5 Algorithm Validation

The reconfiguration decision algorithm must be thoroughly validated in simulation (MATLAB/Simulink) before hardware implementation. Incorrect switching decisions under load could cause short circuits, current spikes, or cell damage. The algorithm uses a rule-based heuristic approach specifically because it is deterministic, verifiable, and predictable — unlike a neural network or reinforcement learning approach, which would be difficult to certify for safety-critical power management.

8.6 Thermal Management of Switching Board

The concentration of 18 MOSFETs on a single PCB generates localized heat, especially during high-current operation. The PCB design must include adequate copper pour for heat spreading, thermal vias, and potentially a small heatsink. This thermal load from the switching board is an additional design consideration that traditional BMS boards do not face.

9. Hardware and Software Requirements

9.1 Hardware Bill of Materials

Category	Item	Est. Cost (INR)
Battery	6x Samsung INR21700-50E cells	~1,800
Battery	21700 cell holders	~300
Switching	18x N-MOSFET (low RDS(on))	~2,700
Switching	9x Gate driver ICs	~900
Sensing	3x INA226 current/voltage sensor	~450
Sensing	6x NTC 10K thermistor	~120
Control	1x STM32F446 development board	~800
Control	1x CAN transceiver module	~150
PCB	Custom PCB fabrication (2-layer)	~500–1,000
Rover	4WD/6WD chassis with DC motors	~1,500–2,500
Rover	Motor driver module (L298N or BTS7960)	~300–500
Misc	Connectors, wires, heatsinks, enclosure	~500
	Estimated Total	~10,000–12,000

Table 8: Estimated bill of materials

9.2 Software Tools

Purpose	Tool
PCB Design	KiCad (open source) or Altium Designer
Firmware Development	STM32CubeIDE / PlatformIO with C/C++
Algorithm Simulation	MATLAB/Simulink (pre-hardware validation)
Data Logging & Analysis	Python (matplotlib, pandas)
Version Control	Git + GitHub
Communication Protocol	CAN bus (vehicle) + I2C/SPI (onboard sensors)

Table 9: Software and development tools

10. Deliverables and Validation Strategy

10.1 Project Deliverables

Custom PCB: MOSFET switching array with integrated sensing, gate drivers, and MCU footprint. Designed in KiCad, fabricated as a 2-layer board.

Firmware: Complete embedded firmware implementing the reconfiguration algorithm, EKF-based SOC estimation, all protection routines, and CAN communication with the vehicle controller.

Integrated Rover: 4WD/6WD rover chassis with the reconfigurable BMS installed, wired to motor drivers, and operational for live demonstration.

Simulation Model: MATLAB/Simulink model of the reconfiguration algorithm and battery equivalent circuit for pre-hardware validation and parameter tuning.

Validation Data: Logged experimental data comparing mission runtime, cell utilization uniformity, thermal performance, and fault resilience between the reconfigurable BMS and a baseline fixed-topology configuration.

Technical Report: Complete design documentation including schematics, PCB layout, firmware architecture, test results, and analysis.

Conference Paper: Potential IEEE conference submission documenting the novel contribution.

10.2 Validation Methodology

The system will be validated through a controlled comparison test. Two identical rover runs will be performed: one with the reconfigurable BMS active and one with it locked in a fixed 3S2P configuration (simulating a traditional BMS). Both runs will use the same pre-conditioned battery pack with a deliberately introduced 10–15% SOC imbalance between cells. The metrics compared will include: total mission runtime (seconds), total energy extracted (Wh), maximum cell temperature during operation (°C), SOC spread at end of mission (%), and number of fault events survived without mission termination.

11. Conclusion and Future Scope

11.1 Conclusion

This project proposes the design and implementation of a mission-adaptive reconfigurable battery management system for an autonomous rover — a system that goes beyond traditional monitor-and-protect BMS architectures by actively reconfiguring the battery pack topology in real time based on SOC and SOA inputs. The three-switch-per-cell topology enables series, parallel, bypass, and hybrid configurations, providing fault tolerance, wear leveling, adaptive current sharing, and maximum energy extraction from every cell in the pack.

The system addresses the fundamental limitations of traditional BMS — stranded energy due to the weakest cell effect, binary fault response, inability to adapt to load conditions, and wasteful cell balancing — with a unified reconfigurable architecture that trades approximately 0.5W of operational overhead for 15–30% more usable energy and fault-resilient operation.

11.2 Future Scope

Module-Level Scaling: Replace per-cell switching with per-module switching for larger battery packs (e.g., 48V home energy storage or EV applications), validating the same algorithms at a different granularity.

Machine Learning Integration: Replace or augment the rule-based decision engine with a reinforcement learning agent trained on historical mission data to optimize reconfiguration strategies for specific mission profiles.

Second-Life Battery Support: Apply the reconfigurable architecture to packs assembled from retired cells with significant parameter heterogeneity, where reconfiguration provides the greatest benefit.

Multi-Platform Validation: Port the BMS to drone and AUV platforms to validate performance across different load profiles, environmental conditions, and safety requirements.

Wireless BMS Communication: Replace wired CAN bus with wireless BMS (wBMS) for improved modularity and reduced wiring complexity in scaled systems.

Digital Twin Integration: Develop a real-time digital twin of the battery pack that runs alongside the physical system, predicting degradation trajectories and optimizing long-term reconfiguration strategies.

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